Retrieval-Augmented Generation (RAG) Explained

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Introduction to RAG

Retrieval-Augmented Generation, usually shortened to RAG, is a technique that makes a large language model (LLM) answer questions using information retrieved from an external knowledge source, such as a set of PDF documents, a company database or a website. Instead of relying only on what the model memorised during training, a RAG system first searches for the most relevant passages and then gives those passages to the model together with the user's question. The model then writes an answer grounded in the retrieved text. RAG was introduced in a 2020 research paper titled "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks" by Patrick Lewis and colleagues at Facebook AI Research (now Meta AI), University College London and New York University. Today RAG is one of the most widely used patterns for building practical AI assistants.

What Problem Does RAG Solve?

RAG solves several problems of plain language models. First, it reduces hallucination, which is when a model confidently invents facts, because the model is told to answer from supplied documents. Second, it fixes outdated knowledge: an LLM only knows information up to its training cut-off date, but a RAG system can read documents updated today. Third, it lets a model use private or domain-specific data, such as a company's policies, a college's rules or a collection of court judgments, that was never part of its training. Fourth, it allows answers to be traced to sources, so users can verify them. Finally, it is cheaper and faster than retraining or fine-tuning a model every time the knowledge changes; you simply update the documents. In short, RAG solves the problems of hallucination, stale knowledge and lack of access to private data.

Embeddings

An embedding is a list of numbers, called a vector, that represents the meaning of a piece of text. An embedding model converts a word, sentence or paragraph into a vector, typically with hundreds or thousands of dimensions, for example 384, 768 or 1536 numbers. The key idea is that texts with similar meaning get vectors that lie close together in vector space, even if they use different words. For example, the sentences "How do I reset my password?" and "I forgot my login credentials" have very similar embeddings. Because of this, a computer can search by meaning rather than only by exact keywords. Popular embedding models include sentence-transformers models such as all-MiniLM-L6-v2, OpenAI's text-embedding models and multilingual models that handle Indian languages.

What Is a Vector Database?

A vector database is a specialised database that stores embeddings and quickly finds the ones most similar to a query embedding. Ordinary databases search for exact matches in rows and columns, but a vector database performs similarity search, also called nearest-neighbour search. Similarity between vectors is usually measured with cosine similarity or Euclidean distance. To stay fast with millions of vectors, vector databases use approximate nearest-neighbour algorithms such as HNSW. Well-known vector databases and libraries include FAISS (from Meta), Chroma, Pinecone, Weaviate, Milvus and Qdrant, and many traditional databases such as PostgreSQL with pgvector now support vectors too. In a small project such as a classroom notebook, an in-memory store like FAISS or Chroma is enough.

How a RAG Pipeline Works

A RAG system has two phases. In the indexing phase, documents are loaded, then split into smaller pieces called chunks, typically a few hundred words each, often with a small overlap so that sentences are not cut off from their context. Each chunk is converted to an embedding and stored in the vector database along with the original text. In the query phase, the user asks a question; the question is converted to an embedding using the same model; the vector database returns the top-k most similar chunks (for example the top three or five); these chunks are inserted into a prompt such as "Answer the question using only the context below. If the answer is not in the context, say you do not know"; and the LLM generates the final answer. This final instruction is important because it makes the bot refuse to guess when the document does not contain the answer.

Quality Factors and Variations

The quality of a RAG system depends on chunk size, the embedding model, the number of chunks retrieved, the quality of the source documents and the prompt. Documents should contain selectable text, because scanned images need OCR first. Improvements include hybrid search, which combines keyword search (BM25) with vector search, re-ranking models that reorder retrieved chunks, metadata filters and query rewriting. Advanced forms include agentic RAG, where an AI agent decides when and what to retrieve, and graph RAG, which uses knowledge graphs. RAG is evaluated by checking whether the right chunks were retrieved and whether the answer is faithful to them.

Uses and Indian Examples

RAG is used in customer-support bots, enterprise search, student helpdesks, medical literature assistants and legal research tools. In India, a popular use is question answering over court judgments, where lawyers search long Supreme Court and High Court documents in plain language. Government service chatbots answering questions about schemes, banks answering policy questions, and college assistants answering admission queries are other examples. Indian startups and training organisations in Chennai teach students to build RAG chatbots using free Colab notebooks.

Challenges and Future

RAG can still fail if retrieval returns the wrong chunks, if documents are poorly formatted, or if the model ignores the context. Long documents need careful chunking, and sensitive data needs access control. The future of RAG includes multimodal retrieval over images and tables, longer context windows, agentic workflows and better evaluation tools.

Quick Facts

- RAG stands for Retrieval-Augmented Generation and was introduced in a 2020 paper by Patrick Lewis and colleagues.
- RAG reduces hallucination, supplies up-to-date knowledge and enables use of private documents.
- Embeddings are numerical vectors that capture the meaning of text.
- A vector database stores embeddings and finds the most similar ones to a query.
- Documents are split into chunks, embedded, retrieved by similarity and passed to an LLM to generate the answer.